Q1 A coil with coil resistance of $8\ \Omega$ and inductance of $0.2\ H$ in a fan is connected to a $240\ V$, $50\ Hz$ a.c. source. Determine,

i) The load impedance

ii) The coil current

iii) The apparent power in complex form

iv) The circuit phase angle

v) The power factor

vi) The true and reactive power

Draw the power triangle and evaluate the power factor of the coil.

Q2 A single-phase motor takes $8.3\ A$ at a power factor of 0.6 lagging when connected to a $230\ V$, $50\ Hz$ supply. Design a power factor correction circuitry to raise the power factor to 0.95.

Q3 A factory has the following loads connected to the mains supply of $415\ V$, $50\ Hz$:

(a) $40\ kVA$, 0.75 lagging p.f.;

(b) $5\ kVA$, unity p.f.;

(c) $10\ kVA$, 0.9 leading p.f.

Find the capacitance needed to correct the power factor to 0.9 lagging.

Q4 A series circuit comprises an inductor, of resistance $10\ \Omega$ and inductance $159\ \mu H$, and a variable capacitor connected to a $50\ mV$ sinusoidal supply of frequency $1\ MHz$.

i) What value of capacitance will result in resonant conditions?

ii) What will then be the current and its magnification?

iii) Determine the bandwidth and the half-power cutoff points.

iv) Calculate the voltage developed across the inductor and capacitor.

Q5 An e.m.f. whose instantaneous value at time t is given by $283\sin(314t + \pi/4)$ volts is applied to an inductive circuit and the current in the circuit is $5.66\sin(314t - \pi/6)$ amperes. Determine:

i) The frequency of the e.m.f.

ii) The resistance and inductance of the circuit

iii) The active power absorbed.

If series capacitance is added to bring the circuit into resonance at this frequency and the above e.m.f. is applied to the resonant circuit, find the corresponding expression for the instantaneous value of the current. Sketch a phasor diagram for this condition. Explain why it is possible to have a much higher voltage across a capacitor than the supply voltage in a series circuit.

Q6 A coil has resistance of $400\ \Omega$ and inductance of $318\ \mu H$. Find the capacitance of a capacitor which, when connected in parallel with the coil, will produce resonance with a supply frequency of $1\ MHz$. If a second capacitor of capacitance $23.5\ pF$ is connected in parallel with the first capacitor, find the frequency at which resonance will occur and its Q-factor.

End